

The non-identity reading in Slavic ATB constructions*

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Abstract

This paper presents Across-the-Board (ATB) constructions with the non-identity reading in Slavic languages that have not received much attention so far. It shows that the data pose a problem for most current approaches, not only for approaches designed to derive the identity reading of ATB constructions but also for approaches that assume the existence of the non-identity interpretation. The article proposes an analysis with multiple movement and a distinctness requirement on linearization in ATB constructions. It also identifies several factors relevant to determining the type of the reading of the moved element. Mostly, the factors can be subsumed under definiteness effects.

Keywords: ATB constructions, multiple movement, non-identity reading, definiteness, distinctness, Slavic

1. The non-identity reading in various ATB constructions

It is a widespread opinion that ATB constructions receive a single identity reading; see e.g. Citko (2005), Reich (2009), Zhang (2009, 2010) and Salzmann (2012). The Slavic data presented in this article – mostly from Russian, Polish and Czech – however show that ATB constructions permit the non-identity reading (also called *respective reading*). This observation is not novel (see e.g. Munn 1999, Gawron & Kehler 2004 and De Vries 2017) but I will add new empirical evidence and will identify several factors that play a role in determining the type of the reading of the moved element.

The non-identity reading can be found in all possible ATB constructions in Slavic: ATB topicalizations, ATB questions (with both arguments and adjuncts), relative clause ATB constructions and left-branch-extraction ATB constructions. For instance, the ATB topicalization below shows that besides the *type identity interpretation* in (1a) (identically entitled book) and the *token identity interpretation* (the same copy of the book) in (1b), the *non-identity reading* is also possible; consider (1c), with the meaning (1d).¹

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1 Example (1c) can also have the type identity interpretation and marginally, also the token identity reading if it holds that Jirka sold the book bought on the internet to a bookstore in which then Marie bought it.

I use the following abbreviations: ACC=accusative, CL=clitic, F=feminine, GEN=genitive, M=masculine, N=neuter, NOM=nominative, PL=plural.

- (1) a. Tu knížku si Jirka objednal přes internet a
 the book.ACC self Jirka.NOM ordered over internet and
 Jana koupila v knihkupectví.
 Jana.NOM bought in bookstore
 ‘The book, Jirka ordered on the internet and Jana bought in a bookstore.’
- b. Tu knížku Jana koupila přes internet a její
 the book.ACC Jana.NOM bought over internet and her
 manžel prodal do knihkupectví.
 husband.NOM sold to bookstore
 ‘The book, Jana bought on the internet and her husband sold to a bookstore.’
- c. Nějakou knížku si Jirka objednal přes internet a
 some book.ACC self Jirka.NOM ordered over internet and
 Jana koupila v knihkupectví.
 Jana.NOM bought in bookstore
 ‘Some book, Jirka ordered on the internet and Jana bought in a bookstore.’
 (Czech)
- d. $\exists x \exists y$. Jirka-bought-on-the-internet (x) & Jana-bought-in-a-bookstore (y)

ATB questions also permit the non-identity reading; consider the Czech example (2a), which is ambiguous between the identity and non-identity reading. The non-identity reading can be controlled by adverbials, as shown in (2b). Consequently, using the definite term instead of the wh-phrase will produce a pragmatically odd sentence, as shown in (2c). Given that finishing a book precedes the review process under normal circumstances, the ordering of the events in the conjunction is peculiar.

- (2) a. Jakou knížku Jana dopsala a Jirka recenzoval?
 what book.ACC Jana.NOM finished and Jirka.NOM reviewed
 ‘What book did Jana finish and Jirka review?’
- b. Jakou knížku Jana dneska dopsala a Jirka
 what book.ACC Jana.NOM today finished and Jirka.NOM
 minulý týden recenzoval?
 last week reviewed
 ‘What book did Jana finish today and Jirka review last week?’
- c. #Tu knížku Jana dneska dopsala a Jirka minulý
 the book.ACC Jana.NOM today finished and Jirka.NOM last
 týden recenzoval.
 week reviewed
 Intended: ‘The book, Jana finished today and Jirka reviewed last week.’
 (Czech)

The non-identity reading can also occur in sentences with *wh*-pronouns. The question in (3) is appropriate, for instance, in a context where two friends, Joanna and Jerzy, went to a flea market and bought and sold two different things.

- (3) Co Joanna kupiła i Jerzy sprzedał?
 what Joanna.NOM bought and Jerzy.NOM sold
 ‘What did Joanna buy and Jerzy sell?’ (Polish)

In relative clause ATB constructions like (4a) the identity reading arises, but if the appropriate noun is pluralized, as in example (4b), the non-identity reading becomes possible.

- (4) a. Ja znaju tu knigu, ktoruju Jana publikowała
 I.NOM know the book.ACC which.ACC Jana.NOM published
 i Jurij recenzował.
 and Jurij.NOM reviewed
 ‘I know the book that Jana published and Jurij reviewed.’
 b. Ja znaju knigi, kotorye Jana publikowała
 I.NOM know books.ACC.PL which.ACC.PL Jana.NOM published
 i Jurij recenzował.
 and Jurij.NOM reviewed
 ‘I know books that Jana published and Jurij reviewed.’ (Russian)

So far, only arguments have been discussed but the non-identity reading is also possible with adjuncts, as shown by example (5a), which has meaning (5b) and can be answered by (5c) (see also Munn 1999). Since between finishing a book manuscript and its publishing, a certain time elapses, (5a) cannot be a question about an identical time point. (However, that does not mean that the question cannot be answered by an identical time specification if the specified time interval is long enough to include both events expressed in (5a)).

- (5) a. Kiedy tę książkę Joanna napisała a wydawnictwo
 when the book.ACC Joanna.NOM wrote and publisher.NOM
 opublikowało?
 published
 ‘When did Joanna finish the book and the publisher publish it?’
 b. At what time x, Joanna finished her book at x and at what time y, the
 publisher published the book at y?
 c. Joanna tę książkę napisała w 2013 roku a
 Joanna.NOM the book.ACC wrote in 2013 year and
 wydawnictwo ją opublikowało w 2015 roku.
 publisher.NOM it.ACC published in 2015 year.
 ‘Joanna finished her book in 2013 and it came out in 2015.’ (Polish)

Generally, for the identity or the non-identity reading to be possible, there should be a situation compatible with the eventualities and referents introduced in the (two) conjuncts. For instance, under the standard background knowledge, the sentence *When did Karl Marx write “Das Kapital” and Mohammed die?*² cannot have the identity reading. At the same time, since it is probably impossible to find a situation for which asking this question makes sense, the non-identity reading seems to be also excluded.

The remainder of the article is structured as follows. Section 2 briefly overviews factors relevant to determining the type of the reading of the moved element. Section 3 discusses shortcomings of previous approaches to ATB constructions. Section 4 proposes an analysis based on multiple movement and a distinctness requirement on linearization. Section 5 concludes the article.

2. Relevant factors

We already saw in the preceding section that the form of the moved element plays an important role in determining the type of the reading – identical versus non-identical –; consider again the contrast between the definite DP in (1a), (1b), (2c) and the indefinite DP in (1c), (2a), and the contrast between the singular, definite DP and the plural, non-quantized DP in (4) (see also De Vries 2017).^{3,4}

In connection with definiteness and quantization, morphological aspect is also relevant, as it has been argued that the perfective aspect makes the verbal object definite or quantized; see e.g. Krifka (1989, 1992), Piñón (1994) and Filip (1999). Specifically, when the imperfective verb *recenzoval* ‘reviewed’ in the ambiguous example (2a) is replaced with the perfective verb *zrecenzoval*, the identity reading becomes strongly preferred, as demonstrated by the example in (6). (Notice, however, that perfectivity is not sufficient condition for the identity reading, as shown by the perfective verbs in (3).)

- (6) Jakou knížku Jana dopsala a Jirka zrecenzoval?
 which book.ACC Jana.NOM finished and Jirka.NOM reviewed
 ‘Which book did Jana finish and Jirka review?’

(Czech)

The presence or absence of a quantifier in the DP also has an impact on the type of the interpretation; compare (4) with the following example, containing *vše* ‘all’.

2 Thanks go to an anonymous reviewer for this example.

3 If wh-phrases are treated as indefinites (e.g. Karttunen 1977).

4 Although North Slavic languages do not have articles, they can use demonstratives to mark definiteness, as in (1a), (1b) and (4a).

- (7) Ja znaju vse knigi, kotorye Jana
 I.NOM know all books.ACC which.ACC Jana.NOM
 publikovala i Jurij recenziroval.
 published and Jurij.NOM reviewed
 ‘I know all books that Jana published and Jurij reviewed.’ (Russian)

In (7), the identity reading is more salient than the non-identity reading, which can be caused by the fact that *vse* is a strong, universal quantifier, whose quantification domain is restricted by contextual pragmatic factors. That is, there is a presupposed set of books published by Jana and reviewed by Jurij and the speaker knows all of them.

If a partitive *wh*-phrase is used, as in example (8), where the questioned referent is chosen from a list of several different books, the non-identity reading is easily available, besides the type identity reading.

- (8) Kterou z těch knih Jana napsala a Jirka
 which.ACC of the books Jana.NOM wrote and Jirka.NOM
 recenzoval?
 reviewed
 ‘Which of the books Jana published and Jirka reviewed?’ (Czech)

Further relevant factors are the meaning of the verb and other elements in the sentence and our world knowledge. For instance, example (9a) is ambiguous between the identity and non-identity reading. In contrast, (9b) seems to allow only the identity interpretation – preferably the token identity reading – because the adverb brings about a closer relation between the two events, hence also between the two objects (which makes the sentence more coherent).⁵ However, if the adverb *předtím* ‘before’ is used, as in (9c), then, given our world knowledge, the most salient reading is the type identity reading; and marginally, the non-identity reading is possible.

- (9) a. Co Pavel koupil a Jana zničila?
 what.ACC Pavel.NOM bought and Jana.NOM destroyed
 ‘What did Pavel buy and Jana destroy?’
 b. Co Pavel koupil a Jana potom zničila?
 what.ACC Pavel.NOM bought and Jana.NOM then destroyed
 ‘What did Pavel buy and Jana then destroy?’
 c. Co Pavel koupil a Jana předtím zničila?
 what.ACC Pavel.NOM bought and Jana.NOM before destroyed
 ‘What did Pavel buy and Jana destroy before?’ (Czech)

5 A weaker effect can be observed if the second proper name is replaced with a relational noun related to participants of the event of the first conjunct, e.g. when *Jana* is replaced with *jeho dcera* ‘his daughter’ in (9a).

Even stronger is the world knowledge effect in example (10a), in which *zničila* ‘destroyed’ is replaced with *spálila* ‘burned’. The order of the conjuncts also plays an important role since sentences are usually temporally interpreted in accordance with the linear order of the expressions describing appropriate events. Given this and our world knowledge, in (10b) – in which the reversed order of the conjuncts fulfills the same function as the adverb *předtím* ‘before’ in (10a) – the token identity reading is strongly dispreferred.

- (10) a. Co Pavel koupil a Jana předtím spálila?
 what.ACC Pavel.NOM bought and Jana.NOM before burned
 ‘What did Pavel buy and Jana burn before?’
 b. Co Jana spálila a Pavel koupil?
 what.ACC Jana.NOM burned and Pavel.NOM bought
 ‘What did Jana burn and Pavel buy?’ (Czech)

In the same vein, in the ATB question in (11), the most salient reading is the non-identity reading since the questioned referent cannot be shared by the two events (the type identity reading is marginally possible with substances like apple puree).

- (11) Co Jerzy na tej imprezie wypił a Paweł
 what.ACC Jerzy.NOM on the party drank and Paweł.NOM
 zjadł?
 ate
 ‘What did Jerzy drink and Paweł eat at the party?’ (Polish)

Another factor affecting the type of the reading of the moved element is binding. The Polish example in (12a), with the pronoun *jego* ‘his’, has the identity reading with one specific teacher given the topicalized status of the moved object (see section 4). In contrast, example (12b), with the anaphor *swojego* ‘one’s’, has the identity reading as well as the non-identity reading.

- (12) a. Jego nauczyciela Jerzy lubi a Joanna nienawidzi.
 his teacher.ACC Jerzy.NOM likes and Joanna.NOM hates
 ‘His teacher, Jerzy likes and Joanna hates.’
 b. Swojego nauczyciela Jerzy lubi a Joanna nienawidzi.
 one’s teacher.ACC Jerzy.NOM likes and Joanna.NOM hates
 ‘His(/her) teacher, Jerzy likes and Joanna hates.’ (Polish)

Information structure of the appropriate sentence is also important. For instance, (13a) allows the identity reading as well as the non-identity reading but (13b) can only receive the two – type and token – identity readings. This probably results from the fact that the existence of the events denoted by the moved, backgrounded verbs is presupposed (Kučerová 2007; Biskup 2011).

- To summarize, the data under discussion show that the non-identity reading is not a rare phenomenon and that the type of the reading depends on syntactic, (lexical) semantic and pragmatic factors.

(15) a. In what city did Mary vacation and Bill decide to live?
b. # Mary vacationed in Paris and Bill decided to live in Toronto.
(Gawron & Kehler 2003: 92)

(16) a. In what cities did Mary vacation and Bill decide to live?

b. Mary vacationed in Paris and Bill decided to live in Toronto.

(Gawron & Kehler 2003: 92)

That means that the unambiguously singular cases like (1c), (2a), (8) and (13) pose a problem for their respective operator approach.

What is more, the Czech translation of the English singular (15), as shown in (17), is fully acceptable, e.g. in the context where Mary and Bill are long time friends and have the same opinion on many topics but with respect to cities, their opinions differ widely.

(17) a. V jakém městě Mary trávila dovolenou a Bill se
in what city Mary spent vacation and Bill self
rozhodl žít?
decided to.live

‘In what city did Mary vacation and Bill decide to live?’

b. Mary trávila dovolenou v Paříži a Bill se rozhodl
Mary spent vacation in Paris and Bill self decided
žít v Torontu.
to.live in Toronto

‘Mary vacationed in Paris and Bill decided to live in Toronto.’

(Czech)

In contrast, the Czech translation of the English plural example (16) is not acceptable, as shown in (18), and the question can only be answered with more cities for each conjunct, as demonstrated in (19).

(18) a. V jakých městech Mary trávila dovolenou a Bill se
in what cities Mary spent vacation and Bill self
rozhodl žít?
decided to.live

In what cities did Mary vacation and Bill decide to live?

b. # Mary trávila dovolenou v Paříži a Bill se rozhodl
Mary spent vacation in Paris and Bill self decided
žít v Torontu.
to.live in Toronto

‘Mary vacationed in Paris and Bill decided to live in Toronto.’

(Czech)

(19) Mary trávila dovolenou v Paříži a Vídni a Bill se
Mary spent vacation in Paris and Vienna and Bill self
rozhodl žít v Torontu a Seattlu.
decided to.live in Toronto and Seattle

‘Mary vacationed in Paris and Vienna and Bill decided to live in Toronto and Seattle.’

(Czech)

Thus, compared with the English answers, Czech behaves in the opposite way. The plural *wh*-phrase cannot be distributed over the two conjuncts in (18). Each conjunct needs to have its own plural *wh*-phrase.

It is also possible to construct examples that are problematic for both approaches (Munn 1999 and Gawron & Kehler 2003, 2004); consider (20b), with a VP coordination and a singular *wh*-phrase. The appropriate context is provided in (20a).

- (20) a. Jirka is traveling around the world and his intention is to visit one capital on each continent. He already was in North America and now he is going to South America.
 b. Jaké hlavní město Jirka navštívil v Severní Americe
 what capital city.ACC Jirka.NOM visited in North America
 a navštíví v Jižní Americe?
 and will.visit in South America
 ‘Which capital from North America did Jirka visit and which capital from South America will he visit?’ (Czech)

Larson (2013) proposes an analysis of the non-identity reading in Macedonian ATB constructions like (21).

- (21) Kakva kniga pročitā Ana i (ja) kupi Ivan?
 what book read Ana and CL bought Ivan
 ‘What book did Ana read and Ivan buy?’

(Macedonian, Larson 2013: 429)

In his analysis, movement to Spec,CP only happens from the first conjunct and in the subsequent conjunct(s), there is literally nothing (the appropriate element is later recovered). This approach has a problem with the fact that there is evidence for movement from the non-initial conjunct, as discussed below (see also arguments in Georgi *et al.* to appear). For instance, recall the Polish example (12b), with the anaphor *swojego* ‘one’s’, for convenience repeated as (22). To be bound, *swojego* must occur in a position c-commanded by the subject. Since *swojego nauczyciela* ‘his teacher’ can have the non-identity reading, it must be able to reconstruct into both conjuncts.

- (22) Swojego nauczyciela Jerzy lubi a Joanna nienawidzi.
 one’s teacher.ACC Jerzy.NOM likes and Joanna.NOM hates
 ‘His(/her) teacher, Jerzy likes and Joanna hates.’ (Polish)

Similarly, the Czech example in (23) demonstrates that *flintu do žita* ‘a gun to the rye’ reconstructs into both conjuncts for the idiomatic interpretation.

- (23) Flintu do žita tady nikdo ještě nehodil a
 gun.ACC to rye here nobody.NOM yet not.threw and
 nikdo ani nehodí.
 nobody.NOM nor notthrows
 ‘Nobody gave up hope here yet and even nobody will.’ (Czech)

The following ungrammatical examples show that strong crossover effects are also present in both conjuncts.

- (24) a. * Č'ju₁ mat' on₁ ne poprivetsvoval a Oleg
 whose mother.ACC er not greeted and Oleg.NOM
 nepoceloval?
 not kissed
- b. * Č'ju₁ mat' Oleg nepoprivetsvoval a on₁
 whose mother.ACC Oleg.NOM not greeted and he.NOM
 nepoceloval?
 not kissed (Russian)

The next argument is based on Condition C effects; consider the Russian examples in (25), which are ungrammatical regardless of whether the r-expression reconstructs below the coindexed pronoun occurs in the first or in the second conjunct.

- (25) a. * Sestre Pavla₁ on₁ dal knigu a Ivan
 sister.DAT Pavel.GEN he.NOM gave book.ACC and Ivan.NOM
 buket.
 bouquet.ACC
- b. * Sestre Pavla₁ Ivan dal knigu a on₁
 sister.DAT Pavel.GEN Ivan.NOM gave book.ACC and he.NOM
 buket.
 bouquet.ACC (Russian)

Another argument for multiple movement can be found in the following example, which contains a flea market scenario and uses an adjunct island as a diagnostic.

- (26) a. Jana prodala zrcadlo a Jirka měl
 Jana.NOM sold mirror.ACC and Jirka.NOM had
 štěstí, protože koupil staré šachy.
 happiness.ACC because bought old chess.ACC
 'Jana sold a mirror and Jirka was lucky because he bought an old chess set.'
- b. * Co_{1,2} Jana prodala t₁ a Jirka měl
 what.ACC Jana.NOM sold and Jirka.NOM had
 štěstí, protože koupil t₂?
 happiness.ACC because bought
- c. * Co_{1,2} Jirka měl štěstí, protože koupil t₁,
 what.ACC Jirka.NOM had happiness.ACC because bought
 a Jana prodala t₂?
 and Jana.NOM sold (Czech)

Examples (25b) and (25c) demonstrate that the ATB construction is ungrammatical independently of whether the adjunct containing the gap occurs in the first or in the second conjunct.

The movement arguments are also problematic for the ATB analysis of the non-identity reading by te Velde (2002, 2005), in which movement does not happen from subsequent conjuncts. Since in his analysis, the feature identity of the parallel elements in conjuncts plays an important role, he also has a problem with syncretism examples. Specifically, there are ATB constructions with case or gender syncretisms, in which the parallel elements do not have identical formal features, as shown in (27) and (28). The Polish example in (27) shows that sentences with parallel elements marked with distinct cases can be grammatical if the case forms are syncretic.

- (27) Kogo_{1,2} Janek lubi t₁ a Jerzy nienawidzi t₂?
 who.ACC/GEN Janek likes t.ACC and Jerzy hates t.GEN
 ‘Who does Janek like and Jerzy hate?’ (Polish, Borsley 1983: 170)

The left-branch-extraction example in (28) demonstrates the same phenomenon for gender features. If the ending of the extracted wh-word is syncretic, as in (28a) for the masculine and neuter genitive, the sentence is grammatical. This contrasts with ungrammatical examples (28b) and (28c), in which *jakého* is not syncretic for the masculine and feminine genitive.

- (28) a. Jakého₁ se Pavel zbavil [t₁ lehátka] a
 what.GEN.M/N self Pavel.NOM got.rid.of cot.GEN.N and
 Jirka [t₁ stolečku]?
 Jirka.NOM table.GEN.M
 ‘What cot did Pavel get rid of and what table did Jirka get rid of?’
 b. * Jakého₁ se Pavel zbavil [t₁ skříňky] a
 what.GEN.M/N self Pavel.NOM got.rid.of box.GEN.F and
 Jirka [t₁ stolečku]?
 Jirka.NOM table.GEN.M
 c. * Jakého₁ se Pavel zbavil [t₁ stolečku] a
 what.GEN.M/N self Pavel.NOM got.rid.of table.GEN.M and
 Jirka [t₁ skříňky]
 Jirka.NOM box.GEN.F (Czech)

In the ATB analysis by Zhang (2009, 2010), the non-identity reading is licensed by plurality – as in Gawron & Kehler (2003, 2004) – and by non-thematic properties of the left-peripheral element. Therefore, this analysis, too, will have a problem with ATB constructions containing a singular extractee. In addition, the adjunct status of the left-peripheral element is also problematic since there are also ATB constructions with an extracted argument, as shown above.

4. Multiple movement and distinctness

Given the movement diagnostics in the preceding section, I propose to analyze ATB constructions with the non-identity reading in terms of two (or more) independent movements. The moved elements occur in Spec,CP positions and conjoined TPs contain their copies. Thus, two different operators bind their corresponding variables, which derives the interpretation with two different referents, i.e. the non-identity reading, as schematized in (29a). ATB topicalizations look like (29b) and ATB questions like (29c).

- (29) a. $[_{CP} XP \lambda_1 [YP \lambda_2 [C' C [_\&P [_{TP} t_1] [_\& [_{TP} t_2]]]]]]]$
 b. As to x, P(x) and as to y, Q(y)
 c. For which x, P(x) and for which y, Q(y)?

That there are independent movement chains is demonstrated by the ATB topicalization in (30), in which the two conjuncts can differ with respect to reconstruction of the topicalized element.

- (30) [Každé dítě]₁ [jeho rodiče]₁ přivedli do školy a
 every child.ACC his parents.NOM brought to school and
 [nějaká učitelka / dvě učitelky] uvedl-a/y do třídy.
 some teacher.NOM two teachers.NOM ushered in classroom
 ‘Every child was brought to the school by his parents and was ushered
 into the classroom by a teacher/two teachers.’ (Czech)

Specifically, (30) can be interpreted in the way that *každé dítě* ‘every child’ binds the pronominal *jeho rodiče* ‘his parents’ in the first conjunct and at the same time, it can be in the scope of the indefinite *nějaká učitelka* ‘a teacher’ or the numeral *dvě učitelky* ‘two teachers’ in the second conjunct.

The multiple movement analysis is supported by the fact that Slavic languages have overt multiple wh-fronting, as shown by the Russian example (31).⁶

- (31) Kto₁ kogo₂ včera t₁ poceloval t₂?
 who.NOM who.ACC yesterday kissed
 ‘Who kissed who yesterday?’ (Russian)

6 The question arises how the non-identity reading is derived in languages like English. Given that none of the current approaches to ATB constructions can derive all the non-identity examples discussed above, the proposed analysis cannot be abandoned. If Bošković & Franks (2000) are correct and the non-identity reading in English ATB constructions cannot be derived by *covert* multiple movement, then we need an alternative mechanism. That different mechanisms are necessary is also suggested by the interpretational contrast between English, German and Czech plural wh-phrases discussed below.

An analysis with more chains was also proposed by George (1980) and Wilder (1994) but they assume conjoined CPs. In what follows, I argue that the TP conjunction is preferred over the CP conjunction. The first argument is based on behavior of clitics in ATB constructions. Dotlačil (2008) argues that clitic omission like the one in (32) is a result of ATB movement.

- (32) Řekl, že jsem ho zavolał a představil známým.
 he.said that AUX.1SG him.ACC called and introduced friends
 ‘He said that I had called him and had introduced him to friends.’

(Czech, Dotlačil 2008: ex. (3))

The clitics *jsem* and *ho* are omitted in the second conjunct. From the fact that the complementizer *že* occurs in the head C and the ATB-moved *jsem* and *ho* follow it and must escape the two conjuncts is obvious that maximally TPs – and not CPs – can be conjoined.

The second argument is based on the difference between the pair list reading and the single pair reading. It has been argued that the reading depends on (the type of) C (Hagstrom 1998; Rudin 2008). Thus, if ATB constructions had two (conjoined) Cs, the conjuncts could have distinct readings. However, this is not the case; e.g. in (33) either both conjuncts have the pair list reading or they have the single pair reading.

- (33) Co komu Pavel dal a Marie sebrala?
 what.ACC who.DAT Pavel gave and Marie took.away
 ‘What did Pavel give to whom and what did Marie take away from whom?’
 (Czech)

An economy consideration also prefers conjunction of TPs dominated by one CP phrase over conjunction of two complete CPs (see Citko’s 2006 structural economy condition).

It has been argued that if a rule applies into a coordinate structure and affects all conjuncts, then it does not violate the Coordinate Structure Constraint (e.g. Ross 1967; Williams 1977). For this reason, I assume that the moving elements are extracted from conjuncts simultaneously, following parallel movement proposals in the literature (see Hiraiwa 2001; Chomsky 2008; Bošković 2012 and Hein & Murphy 2016). To make this operation conceptually more appealing, it can be based on the simultaneous operation Multiple Agree (Hiraiwa 2001), here applied to all conjuncts.

Since it is *syntactic* movement from all conjuncts, various movement effects are correctly expected (in contrast to analyses like Te Velde 2002, 2005 and Larson 2013); see section 3 again. The movement operation also can extract adjuncts as well as arguments (which contrasts with Zhang 2009, 2010). And the movement also allows the non-identity reading with VP coordinations (contrary to Munn 1999), as shown in (34), a Polish version of the English example (14).

- The movement can, of course, extract plural phrases; recall the discussion of the interpretational contrast between English and Czech plural *wh*-phrases in section 3. Therefore, if in examples like (18a), repeated here as (35a), the plural *wh*-phrase *v jakých městech* ‘in which cities’ moves from each conjunct, we expect more cities for each clause. That this prediction is correct is shown by the answer in (19), repeated as (35b).

- Given the infelicitous answer in (35c), it seems that Slavic languages – in contrast to English – cannot use the respective operator for deriving the non-identity reading in ATB constructions (if Gawron & Kehler’s proposal (2003, 2004) is correct). Instead, they can use multiple movement. The contrast is also observed with plural heads of relatives clauses; consider the English ATB example (36), taken from Gawron & Kehler (2004: 193), which demonstrates the respective reading, according to which Susan is related to Bob, Lyn to John and Mary to Bill.

- In contrast, in the slightly modified Czech example in (37), the respective reading is not available; only the identity interpretation is possible. That is, Bob loves the three sisters; John admires the three sisters and Bill hates them.

- (37) a. Konečně jsem dneska potkal Susan, Lyn a Mary.
 ‘I finally met Susan, Lyn, and Mary today.’
 b. To jsou ty tři sestry, které Bob miluje, John obdivuje
 it are the three sisters that Bob loves, John admires
 a Bill nesnáší.
 and Bill hates.
 ‘They are the three sisters that Bob loves, John admires and Bill
 hates.’ (Czech)

What is interesting is that German patterns with Czech regarding the respective operator and the plural head of relative clauses. Example (38) also has the identity reading, in which all three sisters are loved by Bob, admired by John and hated by Bill.

- (38) a. Heute habe ich endlich Susan, Lyn und Mary getroffen.
 ‘I finally met Susan, Lyn, and Mary today.’
 b. Sie sind die drei Schwestern, die Bob mag, John
 they are the three sisters that Bob loves, John
 bewundert und Bill haßt.
 admires and Bill hates
 ‘They are the three sisters that Bob loves, John admires and Bill
 hates.’ (German)

This suggests that a more fine-grained analysis of ATB constructions – and wh-movement – is necessary (see also footnote 6). On one side, German patterns with English; it lacks overt multiple wh-movement and allows ATB movement. On the other hand, it behaves like Czech and contrasts with English: it has the identity interpretation in ATB constructions with relatives headed by a plural noun.

Moreover, Pankau (2013) shows that German can combine ATB movement with wh-copying; consider the following example.

- (39) a. Wen glaubst du [_{&P} [dass Peter t liebt] und [dass Max t schlägt]]?
 who believe you that Peter loves and that Max beats
 b. Wen glaubst du [_{&P} [wen Peter t liebt] und [wen Max t schlägt]]?
 who believe you who Peter loves and who Max beats
 c. Wen glaubst du [_{&P} [wen [Peter t liebt] und [Max t schlägt]]?
 who believe you who Peter loves and Max beats
 d. *Wen glaubst du [_{&P} [Peter t liebt] und [wen [Max t schlägt]]?
 who believe you Peter loves and who Max beats
 ‘Who do you think that Peter loves and that Max beats?’
 (German, Pankau 2013: 21)

(42) Co na té party budeš jíst a my budeme pít?
what.ACC on the party will.2SG eat and we will.1PL drink
'What will we eat and you drink at the party?' (Czech)

(43) [_{CP} Co [_{C'} komu [_{C'} ee [_{C'} ~~komu~~ C [_{&P} Pavel dal a
what.ACC who.DAT what.ACC who.DAT Pavel gave and
Marie sebrala?]]]]]
Marie took.away
‘What did Pavel give to whom and what did Marie take away from
whom?’ (Czech)

7 Richards' (2010) original proposal would block the Slavic multiple wh-fronting in questions generally.

cannot be distinguished from one another given their identical phonological properties. Therefore, they are treated as a single phrase and the left(most) *co* is linearized. The same reasoning also applies to the parallel wh-phrases *komu* in (43).⁸

It has been argued that wh-phrases in multiple wh-movement can also target different projections in the left periphery in Slavic (see e.g. discussion in Meyer 2003). If it also holds for ATB-movement, then e.g. the wh-phrases in (43) do not form multiple specifiers of the single CP but occur in different projections of the split left periphery (e.g. Rizzi 1997). This does not pose a problem for the current analysis if the landing sites belong to the same CP phase. Notice, however, that the omission must not violate the principle of recoverability (see e.g. Chomsky 1965: 177, 222).⁹

Although the article is primarily concerned with the non-identity reading, let us briefly look at the identity reading, too. It helps to understand why in some cases, the non-identity reading is dispreferred or impossible. In the preceding sections, it was shown that factors affecting the reading of the extracted element are of different nature; they can be syntactic, (lexical) semantic and pragmatic. We observed that in contrast to the non-identity reading, the identity reading has to do with definiteness, presuppositionality and backgrounding, which in turn have to do with the place of interpretation and non-reconstruction. There are two basic options with respect to the identity interpretation. The first possibility is that there are two different mechanisms, e.g. one of the proposals discussed in the beginning of section 3 for the identity reading and then the current proposal for the non-identity reading. The problem with this option is that there would have to be some sort of conspiracy: If one of the relevant factors discussed in section 2 is going to appear in the derivation, then the specific mechanism designed for deriving the identity reading should be used.

The second option is that there is only one mechanism – multiple movement – and the semantic-pragmatic interface decides whether or not the parallel element from the subsequent conjunct(s) will pick up the same referent as the constituent from the first conjunct. Since definite and presuppositional expressions are typically used for referring to backgrounded (known) entities, then if the two movement chains contain a definite or presuppositional element, it will come as no surprise that the element from the second conjunct refers to the referent that was introduced by the

8 As pointed out to me by an anonymous reviewer, the current analysis could account for the following contrast with respect to Postal's (1998) interwoven dependencies.

- (i) Which nurse₁ and which hostess₂ did Fred date t₁ and Bob marry t₂, respectively?
- (ii) * Who₁ and who₂ did Fred date t₁ and Bob marry t₂, respectively?

In contrast to (i), example (ii) is ungrammatical since it contains two phonologically identical wh-phrases in CP, which cannot be linearized.

9 For this reason, e.g. the second *čto* 'what' in the Russian *Čto čto pričinilo?* 'What caused what?' or the second *koj* 'who' in the Bulgarian *Koj koj e?* 'Who is who?' cannot be deleted (but see Roscoe & Thoms (to appear)). The same reasoning would apply to embedding cases in which the two fronted wh-phrases originate from different clauses (the matrix and the embedded one).

parallel element in the first conjunct, as in (1a), (1b) and (4a), with the definite object, and in (7), with the presuppositional quantifier *vse* ‘all’.

In the same vein, if it is correct that the perfective aspect of the verb makes the verbal object definite in the second conjunct in example (6), then the moved expression will again want to pick up the same referent as the constituent extracted from the first conjunct. The referent of the object of the first conjunct is the most salient potential antecedent for the object of the second conjunct.¹⁰

An analogous reasoning can be used for constructions like (12a), with the ATB-topicalized pronoun *jego* ‘his’, since non-deictically used pronouns refer to backgrounded entities as well. This contrasts with example (12b), which has the non-identity reading in addition to the identity reading, given that the ATB-moved anaphoric elements can reconstruct and be bound by the appropriate subjects in accordance with the Condition A.

The case with verb movement in example (13b) is more complex. If the verbs move to the background domain in the two conjuncts and the existence of the events denoted by them is presupposed (Kučerová 2007; Biskup 2011), then a mapping mechanism similar to the case with the perfective verb in (6) can be used. Specifically, the moved, backgrounded verb makes its object definite (at least in the second conjunct in (13b)) and consequently, the *wh*-expression of the second conjunct will refer to the same entity as the *wh*-object of the first conjunct.

As to the question why the identity reading is generally preferred in ATB constructions, one could hypothesize for the second option that the preference is based on the fact that there is a tendency to interpret sentences (discourse) in a coherent manner; see e.g. Kehler (1996, 2002).

5. Conclusions

The Slavic data have shown that the non-identity reading of the overtly moved element in ATB constructions is not exceptional. It can be found in ATB constructions with a relative clause, in ATB topicalizations, ATB questions and left-branch-extraction ATB constructions. The moved element can be an argument as well as an adjunct. The type of the reading – identical versus non-identical – depends on various factors, e.g. definiteness, backgrounding, world knowledge and lexical semantic properties of the verb. It has been shown that the presented ATB data also pose a problem for approaches designed to derive the non-identity reading. In the current approach, the non-identity reading is analyzed in terms of independent movement chains and a distinctness requirement on linearization.

¹⁰ The connection between perfectivity and definiteness could be based on the notion of maximality; see e.g. Filip (2008) for an analysis of perfectivity in terms of a maximalization operator on events on one side and on the other side, Chierchia (1998), according to whom definiteness brings about the reference to the largest singularity or plurality.

References

- Billings, Loren & Rudin, Catherine. 1996. Optimality and superiority: A new approach to overt multiple *wh*-ordering. In Toman, Jindřich (ed.), *Formal Approaches to Slavic Linguistics: The College Park Meeting, 1994*, 35–60. Ann Arbor: Michigan Slavic Publications.
- Biskup, Petr. 2011. *Adverbials and the phase model*. Amsterdam: John Benjamins.
- Biskup, Petr. 2018. Case syncretism in Russian, Polish and Czech ATB constructions. In Browne, Wayne, Despić, Miloje, Enzinna, Naomi, Karlin, Robin, De Lemos, Simone & Zec, Draga (eds.), *Proceedings of FASL 25. The Cornell Meeting 2016*, 418–437. Ann Arbor, MI: Michigan Slavic Publications.
- Bjorkman, Bronwyn. 2014. Accounting for unexpected subject gaps in TP coordination. *The Linguistic Review* 31, 487–513.
- Blümel, Andreas. 2013. *Propagating symmetry*. Frankfurt am Main: Johann Wolfgang Goethe-Universität dissertation.
- Bondaruk, Anna. 2003. Parasitic gaps and ATB in Polish. *Journal of Slavic Linguistics* 11, 221–249.
- Borsley, Robert D. 1983. A note on the Generalized Left Branch Condition. *Linguistic Inquiry* 14, 169–174.
- Bošković, Željko. 2002. On multiple *wh*-fronting. *Linguistic Inquiry* 33, 351–383.
- Bošković, Željko. 2012. Don't feed your movements when you shift your objects. In Uribe-Etxebarria, Myriam & Valmala, Vidal (eds.), *Ways of structure building*, 245–252. Oxford: Oxford University Press.
- Bošković, Željko & Franks, Steven. 2000. Across-the-board movement and LF. *Lingua* 3, 107–129.
- Chierchia, Gennaro. 1998. Reference to kinds across languages. *Natural Language Semantics* 6, 339–405.
- Chomsky, Noam. 1965. *Aspects of the theory of syntax*. Cambridge, MA: MIT Press.
- Chomsky, Noam. 2008. On phases. In Freidin, Robert, Otero, Carlos P. & Zubizarreta, Maria Luisa (eds.), *Foundational issues in linguistic theory: Essays in honor of Jean-Roger Vergnaud*, 133–165. Cambridge, MA: MIT Press.
- Citko, Barbara. 2005. On the nature of Merge: External Merge, Internal Merge, and Parallel Merge. *Linguistic Inquiry* 36, 475–496.
- Citko, Barbara. 2006. The interaction between Across-the-Board *wh*-movement and Left-Branch Extraction. *Syntax* 9, 225–247.
- Dotlačil, Jakub. 2008. Across-the-Board extraction and clitic omission in Czech. In Antonenko, Andrei, Bailyn, John F. & Bethin, Christina Y. (eds.), *Proceedings of Formal Approaches to Slavic Linguistics 16: The Stony Brook Meeting*, 134–150. Ann Arbor, MI: Michigan Slavic Publications.
- Dyła, Stefan. 1984. Across-the-board dependencies and case in Polish. *Linguistic Inquiry* 15, 701–705.
- Filip, Hana. 1999. *Aspect, eventuality types and noun phrase semantics*. New York, London: Garland Publishing.
- Filip, Hana. 2008. Events and Maximalization. In Rothstein, Susan D. (ed.), *Theoretical and Crosslinguistic Approaches to the Semantics of Aspect*, 217–256. Amsterdam: John Benjamins.
- Fox, Danny & Pesetsky, David. 2005. Cyclic linearization of syntactic structure. *Theoretical Linguistics* 31, 1–45.

- Franks, Steven. 1993. On parallelism in across-the-board dependencies. *Linguistic Inquiry* 24, 509–529.
- Franks, Steven. 1995. *Parameters of Slavic morphosyntax*. Oxford: Oxford University Press.
- Gawron, Jean Mark & Kehler, Andrew. 2003. Respective answers to coordinated questions. In Young, Robert B. & Zhou, Yuping (eds), *Proceedings of SALT 13*, 91–108. Ithaca, NY: Cornell University.
- Gawron, Jean Mark & Kehler, Andrew. 2004. The semantics of respective readings, conjunction, and filler-gap dependencies. *Linguistics and Philosophy* 27, 169–207.
- George, Leland M. 1980. *Analogical generalization in natural language syntax*. Cambridge, MA: MIT dissertation.
- Georgi, Doreen, Murphy, Andrew, Szarvas, Timea & Rothert, Johannes. To appear. Diagnosing syntactic structure in ATB and RNR constructions: A reply to Larson (2013, 2014). *Proceedings from Chicago Linguistic Society 60*. University of Chicago.
- Hagstrom, Paul Alan. 1998. *Decomposing Questions*. Cambridge, MA: MIT dissertation.
- Hein, Johannes & Murphy, Andrew. 2016. Case matching and syncretism in ATB dependencies. In Barnickel, Katja, Naranjo, Matías Guzmán, Hein, Johannes, Korsah, Sampson, Murphy, Andrew, Paschen, Ludger, Puškar, Zorica & Zaleska, Joanna (eds.), *Replicative processes in grammar. Linguistische Arbeitsberichte* 93, 301–350. Universität Leipzig.
- Hiraiwa, Ken. 2001. Multiple agree and the defective intervention constraint in Japanese. *MIT Working Papers in Linguistics* 40, 67–80.
- Hornstein, Norbert & Nunes, Jairo. 2002. On asymmetries between parasitic gaps and across-the-board constructions. *Syntax* 5, 26–54.
- Karttunen, Lauri. 1977. Syntax and semantics of questions. *Linguistics and Philosophy* 1, 3–44.
- Kasai, Hironobu. 2004. Two notes on ATB movement. *Language and Linguistics* 5, 167–188.
- Kehler, Andrew. 1996. Coherence and the Coordinate Structure Constraint. In Johnson, Jay, Juge, Matt & Moxley, Jerry (eds.), *Proceedings of the 22nd Annual Meeting of the Berkeley Linguistics Society* 22, 220–231. Berkeley, CA: Berkeley Linguistics Society.
- Kehler, Andrew. 2002. *Coherence, reference, and the theory of grammar*. Stanford, CA: CSLI Publications.
- Krifka, Manfred. 1989. *Nominalreferenz und Zeitkonstitution. Zur Semantik von Massentermen, Pluraltermen und Aspektklassen*. München: Fink.
- Krifka, Manfred. 1992. Thematic relations as links between nominal reference and temporal constitution. In Sag, Ivan A. & Szabolcsi, Anna (eds.), *Lexical Matters*, 29–53. Stanford, CA: CSLI Publications.
- Kučerová, Ivona. 2007. *The syntax of givenness*. Cambridge, MA: MIT dissertation.
- Larson, Bradley T. 2013. Not-so-across-the-board movement in Macedonian. In Franks, Steven, Fowler, George, Dickinson, Marcus, Witcombe, Melissa & Zanon, Ksenia (eds.), *Formal Approaches to Slavic Linguistics 21: The Third Indiana Meeting*, 418–431. Ann Arbor, MI: Michigan Slavic Publications.
- Meyer, Roland. 2003. On multiple wh-fronting and wh-clustering in Czech. In Browne, Wayles, Kim, Ji-Yung, Partee, Barbara H. & Rothstein, Robert A. (eds.), *Proceedings of FASL 11. The Amherst Meeting*, 393–412. Ann Arbor: Michigan Slavic Publications.
- Munn, Alan. 1992. A null operator analysis of ATB gaps. *The Linguistic Review* 9, 1–26.
- Munn, Alan. 1999. On the identity requirement of ATB-extraction. *Natural Language Semantics* 7, 421–425.
- Nunes, Jairo. 2004. *Linearization of chains and sideward movement*. Cambridge, MA: MIT Press.

- Pankau, Andreas. 2013. *Replacing copies. The syntax of wh-copying in German*. Utrecht: LOT publications.
- Piñón, Christopher J. 1994. Accumulation and aspectuality in Polish. In Gonzàles, Mercè (ed.), *Proceedings of the North East Linguistic Society* 24, 491–506. University of Massachusetts, Amherst.
- Postal, Paul M. 1998. *Three investigations of extraction*. Cambridge, MA: MIT Press.
- Reich, Ingo. 2009. *Asymmetrische Koordination im Deutschen*. Tübingen: Stauffenburg.
- Richards, Norvin. 2010. *Uttering trees*. Cambridge, MA: MIT Press.
- Roscoe, Jules & Thoms, Gary. To appear. Anti-anti-homophony: towards a syntactic account of “phonological” *wh*-cluster restrictions in Bulgarian. Paper presented at FASL 2024, Halifax.
- Ross, John R. 1967. Constraints on variables in syntax. Cambridge, MA: MIT dissertation.
- Rudin, Catherine. 2008. Pair-list vs. single pair readings in multiple *wh* free relatives and cor-relatives. *Kansas Working Papers in Linguistics* 30, 257–267.
- Salzmann, Martin. 2012. A derivational ellipsis approach to ATB-movement. *The Linguistic Review* 29, 397–438.
- te Velde, John R. 2002. Phases in the derivation of elliptical coordinate constructions in Germanic. In Abraham, Werner & Zwart, C. Jan-Wouter (eds.), *Issues in formal German(ic) typology*, 307–329. Amsterdam: John Benjamins.
- te Velde, John R. 2005. *Deriving coordinate symmetries*. Amsterdam: John Benjamins.
- de Vries, Mark. 2017. Across-the-board phenomena. In Everaert, Martin & van Riemsdijk, Henk (eds.), *Blackwell Companion to Syntax. 2nd edition*. 1–31. Hoboken: Wiley-Blackwell.
- Wilder, Chris. 1994. Coordination, ATB and ellipsis. In Zwart, C. Jan-Wouter (ed.), *Minimalism and Kayne’s asymmetry hypothesis. Groninger Arbeiten zur germanistischen Linguistik* 37, 291–329. Groningen.
- Williams, Edwin. 1977. Across-the-board application of rules. *Linguistic Inquiry* 8, 419–423.
- Williams, Edwin. 1978. Across-the-board rule application. *Linguistic Inquiry* 9, 31–43.
- Zhang, Niina Ning. 2009. The syntax of same and ATB constructions. *Canadian Journal of Linguistics* 54, 367–399.
- Zhang, Niina Ning. 2010. *Coordination in syntax*. Cambridge: Cambridge University Press.